

RUNNING LED LIGHT

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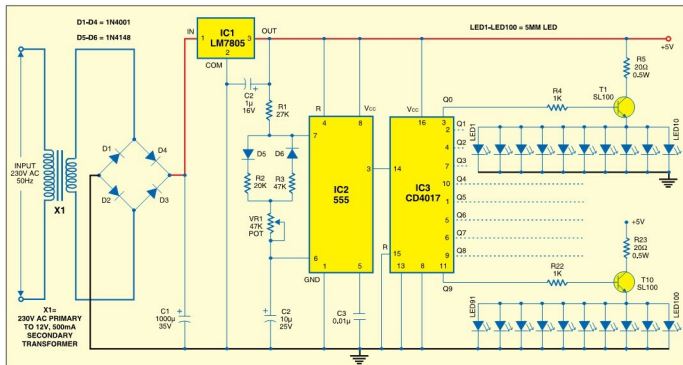
This circuit generates flashing light in running pattern. It uses 100 LEDs in all. The LEDs are arranged in groups of ten. So each group comprises ten LEDs. These ten

bridge rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC1). Regulator 7805 provides regulated 5V for operating the circuit.

Timer IC 555 (IC2) is wired as an astable multivibrator that outputs a



When Q0 goes high, LED-driver transistor T1 conducts and LED1 through LED10 glow. Similarly, when Q1 goes high, LED-driver transistor T2



groups of LEDs light up sequentially.

The circuit uses a bridge rectifier comprising diodes D1 through D4, 5V regulator, timer 555, decade counter CD4017, different-colour 100 LEDs and a few discrete components.

Transformer X1 steps down the mains power to deliver the secondary output of 12V. The transformer output is rectified by the full-wave

square wave of up to 1 Hz. This output of IC2 is fed to clock pin of decade counter IC3 (CD4017) and used for controlling the lighting up sequence of LEDs. The output of IC3 goes high one after another in the sequence of Q0, Q1, Q2, Q9. Thereafter, the cycle repeats from Q0. The frequency or the speed of glowing of LEDs can be controlled by potmeter VR1.

conducts and LED11 through LED20 glow. Thus the groups of ten LEDs glow one after another.

Assemble the circuit on a general-purpose PCB and enclose in a small cabinet. You can use different-coloured LEDs for different groups. Connect the potentiometer on the front panel of the cabinet so that you can easily change the frequency of flashing LEDs. ●